What exactly is a Cayley transform and how to generalize it?

Presenter:
Roberto García Antonio
CIMAT
59th Congress of the Mexican Mathematical Society.

October 23, 2026

Definition 0.1

*A **Lie group** G is a differentiable manifold G equipped with a group structure $(G, \cdot)$ such that the product and the inversion*

$$\cdot : G \times G \to G \qquad i : G \to G$$
$$(g_1, g_2) \mapsto g_1 \cdot g_2 \qquad g \mapsto g^{-1}$$

are smooth functions.

Classical Groups: $\mathrm{SL}(2,\mathbb{R})$ and $\mathrm{SU}(1,1)$

The special linear group $\mathrm{SL}(2,\mathbb{R})$

It is the group of real 2×2 matrices with determinant equal to 1:

$$\mathrm{SL}(2,\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2\times 2}(\mathbb{R}) \;\middle|\; ad - bc = 1 \right\}$$

The group $\mathrm{SU}(1,1)$

It is the group of complex 2×2 matrices with determinant 1 that preserve the Hermitian form of signature $(1,1)$:

$$\mathrm{SU}(1,1) = \left\{ \begin{pmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{pmatrix} \in M_{2\times 2}(\mathbb{C}) \;\middle|\; |\alpha|^2 - |\beta|^2 = 1 \right\}$$

Group actions

Definition 0.2

Let G be a group (not necessarily topological) and X a set. A ***left action*** *of G on X is a function $\varphi : G \times X \to X$, which we denote by $(g, x) \mapsto g \cdot x$, with the following properties:*

- $g_1 \cdot (g_2 \cdot x) = (g_1 g_2) \cdot x \quad (\forall x \in X) \quad$ *and* $\quad (\forall g_1, g_2 \in G)$,
- $1 \cdot x = x \quad (\forall x \in X)$.

Definition 0.3

Let G be a group acting on the left on a space X. We say that the action is ***transitive*** *if for any $x, y \in X$ there exists a $g \in G$ such that*

$$x = g \cdot y.$$

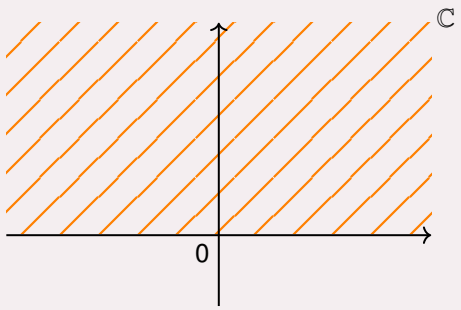
Action of $\mathrm{SL}(2,\mathbb{R})$ on $\mathbb{H}$

Example 1

The group $\mathrm{SL}(2,\mathbb{R})$ acts smoothly on the upper half-plane $\mathbb{H} = \{z \in \mathbb{C} : \mathrm{Im}\ z > 0\}$ by means of Möbius transformations, that is,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot z = \frac{az+b}{cz+d}.$$

This action is transitive.



Action of $\mathrm{SU}(1,1)$ on the unit disk

Möbius transformations

The group $\mathrm{SU}(1,1)$ acts on the unit disk $\mathbb{D} = \{z \in \mathbb{C} \mid |z| < 1\}$ by linear fractional transformations.

For an element $g = \begin{pmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{pmatrix} \in \mathrm{SU}(1,1)$ and a point $z \in \mathbb{D}$, the action is given by:

$$g \cdot z = \frac{\alpha z + \beta}{\bar{\beta} z + \bar{\alpha}}$$

The disk and the half-plane as quotients of Lie groups I

The unit disk $\mathbb{D}$

Let us consider

$$K = \left\{ \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} \middle| \, \theta \in \mathbb{R} \right\} \cong \mathrm{U}(1)$$

We can identify the disk with the homogeneous space:

$$\mathbb{D} \cong \mathrm{SU}(1,1)/\mathrm{U}(1)$$

The disk and the half-plane as quotients of Lie groups II

The upper half-plane $\mathbb{H}$

Let us consider the subgroup

$$K_2 = \left\{ \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \middle| \, \theta \in \mathbb{R} \right\} = \mathrm{SO}(2)$$

We can identify the upper half-plane with the quotient

$$\mathbb{H} \cong \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$$

Bounded symmetric domains - Light version

Definition 0.4

A bounded symmetric domain Ω is an open subset of $\mathbb{C}^n$ of the form G/K, with G a connected semisimple group, such that G acts by biholomorphisms on Ω.

Cayley Transform

Definition 0.5

*The **Cayley transform** is the biholomorphic mapping $\phi : \mathbb{H} \to \mathbb{D}$ from the upper half-plane to the unit disk, defined by:*

$$\phi(z) = \frac{z - i}{z + i}$$

whose inverse transform $\phi^{-1} : \mathbb{D} \to \mathbb{H}$ is given by:

$$z = i\frac{1 + w}{1 - w}$$

How is it generalized?

Definition 0.6 (Jordan Algebra)

*A **Jordan algebra** is a non-associative algebra A over a field (usually $\mathbb{R}$ or $\mathbb{C}$) equipped with a bilinear product $\circ$ that satisfies two axioms for all $x, y \in A$:*

1. ***Commutativity:*** $x \circ y = y \circ x$
2. ***Jordan Identity:*** $(x \circ y) \circ (x \circ x) = x \circ (y \circ (x \circ x))$

Example: Complex self-adjoint matrices with the special product $x \circ y = \frac{1}{2}(xy + yx)$.

Siegel Domains (tube type :D)

Definition 0.7 (Tube Type Domain)

Let $\Omega \subset V$ be a symmetric cone (open, convex, proper and homogeneous). The tube domain is defined as $T_\Omega = V + i\Omega$, where V is a real vector space.

Note that $T_\Omega \subset V^{\mathbb{C}} = V + iV$

In this case T_Ω is a symmetric domain, it is a way to generalize the half-plane. Since $T_{\mathbb{R}^+} = \mathbb{H}$

The mapping p:

$$p(z) = (z - ie)(z + ie)^{-1},$$

is a holomorphic isomorphism from the tube T_Ω onto the domain $\mathcal{D}$.

A bit further

With a bit more Lie algebra theory, we can define a generalized Cayley transform that is no longer valid only for tube type domains. It can be read in [KWU64]

Thank you for your attention :D

[KWU64] A. Korányi, J.A. Wolf, and Berkeley. Department of Mathematics University of California. *Realization of Hermetian Symmetric Spaces as Generalized Half-planes*. Department of Mathematics, University of California, 1964. URL: https://books.google.com.mx/books?id=_6tGHAAACAAJ.

[Con94] J.B. Conway. *A Course in Functional Analysis*. Graduate Texts in Mathematics. Springer New York, 1994. ISBN: 9780387972459. URL: https://books.google.com.mx/books?id=ix4P1e6AkeIC.

[Kna01] A.W. Knapp. *Representation Theory of Semisimple Groups: An Overview Based on Examples*. Princeton landmarks in mathematics and physics. Princeton University Press, 2001. ISBN: 9780691090894. URL: https://books.google.com.mx/books?id=QCcW1h835pwC.

[Lee12] J. Lee. *Introduction to Smooth Manifolds*. Graduate Texts in Mathematics. Springer New York, 2012. ISBN: 9781441999825. URL: https://books.google.com.mx/books?id=xygVcKGPsNwC.